

ON THE TRAIL OF ELUSIVE PARTICLES

Objects on the quantum scale are invisible, but particle detectors, such as cloud chambers and bubble chambers, can make their paths visible.

In a cloud chamber, ionizing particles speed through a vapor-filled chamber, leaving a trail of ionized atoms, around which condensation forms. Within magnetic and electric fields, the condensation forms uniquely curving trails, which allow characteristics such as charge and mass to be calculated.

BUBBLE CHAMBER

Bubble chambers use superheated fluids, and bubbles of gas form around points of ionization to create trails.